

Linear Algebra for AI

Resit Exam 2024

① $T(x) = \begin{bmatrix} x_1 + 4x_2 + 5x_3 + 6x_4 + 9x_5 \\ 3x_1 - 2x_2 + x_3 + 4x_4 - x_5 \\ -x_1 - x_3 - 2x_4 - x_5 \\ 2x_1 + 3x_2 + 5x_3 + 7x_4 + 8x_5 \end{bmatrix}$

a) Standard matrix of T :

$$A = \begin{bmatrix} 1 & 4 & 5 & 6 & 9 \\ 3 & -2 & 1 & 4 & -1 \\ -1 & 0 & -1 & -2 & -1 \\ 2 & 3 & 5 & 7 & 8 \end{bmatrix}$$

b) Basis for $\text{Nul } A$. Is T injective?

To find a basis of $\text{Nul } A$ we find solutions of $Ax = 0$.

$$\left[\begin{array}{ccccc|c} 1 & 4 & 5 & 6 & 9 & 0 \\ 3 & -2 & 1 & 4 & -1 & 0 \\ -1 & 0 & -1 & -2 & -1 & 0 \\ 2 & 3 & 5 & 7 & 8 & 0 \end{array} \right] \sim \left[\begin{array}{ccccc|c} 1 & 4 & 5 & 6 & 9 & 0 \\ 0 & -14 & -14 & -14 & -28 & 0 \\ 0 & 4 & 4 & 4 & 8 & 0 \\ 0 & -5 & -5 & -5 & -10 & 0 \end{array} \right] \sim \left[\begin{array}{ccccc|c} 1 & 4 & 5 & 6 & 9 & 0 \\ 0 & 1 & 1 & 1 & 2 & 0 \\ 0 & 1 & 1 & 1 & 2 & 0 \\ 0 & 1 & 1 & 1 & 2 & 0 \end{array} \right] \sim \left[\begin{array}{ccccc|c} 1 & 0 & 1 & 2 & 1 & 0 \\ 0 & 1 & 1 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$$\Rightarrow \begin{cases} x_1 + x_3 + 2x_4 + x_5 = 0 \\ x_2 + x_3 + x_4 + 2x_5 = 0 \end{cases} \rightarrow \begin{cases} x_1 = -x_3 - 2x_4 - x_5 \\ x_2 = -x_3 - x_4 - 2x_5 \end{cases} \Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = x_3 \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -2 \\ -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -1 \\ -2 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

So a basis for $\text{Nul } A$ is $\left\{ \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ -2 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$. No, the map is not one-to-one (injective) since there are non-trivial solutions to $Ax = 0$.

c) Find a basis for $\text{Col } A$. Is the map surjective?

We need to find the pivot columns of A for that. We can use the row reduction from before:

$$A \sim \left[\begin{array}{ccccc} 1 & 0 & 1 & 2 & 1 \\ 0 & 1 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right], \text{ which tells us that the first two columns are the pivot columns, so a basis for } \text{Col } A \text{ is } \left\{ \begin{bmatrix} 1 \\ 3 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 4 \\ -2 \\ 0 \\ 3 \end{bmatrix} \right\}$$

The map is not surjective since the dimension of the column space is 2, whereas the dimension of the codomain is 4. In other words, the rank of A is less than its number of rows.

② $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & k \\ -6 & 4k & 2 \end{bmatrix}, k \in \mathbb{R}, b = \begin{bmatrix} 1 \\ 1 \\ -3 \end{bmatrix}$

a) Solve $Ax = b$ by using Cramer's rule.

Using Cramer's rule formulas we get $\det A = \begin{vmatrix} 2 & k \\ 4k & 2 \end{vmatrix} + \begin{vmatrix} 0 & 2 \\ -6 & 4k \end{vmatrix} = 4 - 4k^2 + 12 = -4k^2 + 16$

$$\det(A_x(b)) = \begin{vmatrix} 2 & k \\ 4k & 2 \end{vmatrix} + \begin{vmatrix} 1 & 2 \\ -3 & 4k \end{vmatrix} = 4 - 4k^2 + 4k + 6 = -4k^2 + 4k + 10$$

$$\det(A_{x_2}(b)) = \begin{vmatrix} 1 & k \\ -3 & 2 \end{vmatrix} - 6 \begin{vmatrix} 1 & 1 \\ 1 & k \end{vmatrix} = 2 + 3k - 6(k-1) = 2 + 3k - 6k + 6 = -3k + 8$$

$$\det(A_{x_3}(b)) = \begin{vmatrix} 2 & 1 \\ 4k & -3 \end{vmatrix} + \begin{vmatrix} 0 & 2 \\ -6 & 4k \end{vmatrix} = -6 - 4k + 12 = -4k + 6$$

$$x_1 = \frac{\det(A_{x_1}(b))}{\det A} = \frac{-4k^2 + 4k + 10}{-4k^2 + 16}, \quad x_2 = \frac{\det(A_{x_2}(b))}{\det A} = \frac{-3k + 8}{-4k^2 + 16}, \quad x_3 = \frac{\det(A_{x_3}(b))}{\det A} = \frac{-4k + 6}{-4k^2 + 16}$$

b) For which k is A invertible?

A will be invertible iff $\det A \neq 0$, so $\det A = -4k^2 + 16 = 0 \rightarrow k = \pm 2$, so A is invertible $\forall k \in \mathbb{R} \setminus \{\pm 2\}$.
(any k except ± 2)

c) For $k=1$ find A^{-1} .

We know A is invertible for $k=1$, so $[A | Id] \sim [Id | A^{-1}]$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ -6 & 4 & 2 & 0 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 0 & 4 & 8 & 6 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 6 & 6 & -2 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & -1/3 & -1/6 \\ 0 & 1 & 0 & -1/2 & 2/3 & -1/12 \\ 0 & 0 & 1 & 1 & -1/3 & 1/6 \end{array} \right] = [Id | A^{-1}]$$

③ Consider $\mathcal{U} = \{p \in \mathbb{P}_n : p(z) = 0\} \subset \mathbb{P}_n$.

a) Show that $\mathcal{U}$ is a subspace of $\mathbb{P}_n$.

For that we need to prove 3 things:

- $0 \in \mathcal{U}$. The zero polynomial is $p(t) = 0 \forall t$, so clearly $p(z) = 0$ and $0 \in \mathcal{U}$.

- $p, q \in \mathcal{U} \rightarrow p+q \in \mathcal{U}$. So let $p, q \in \mathcal{U}$, then $p(z) = 0, q(z) = 0$. Then $(p+q)(t) = p(t) + q(t)$ (because $a_0 + b_0 + (a_1 + b_1)t + (a_2 + b_2)t^2 + \dots + (a_n + b_n)t^n = a_0 + a_1t + \dots + a_nt^n + b_0 + b_1t + \dots + b_nt^n$), so $(p+q)(z) = p(z) + q(z) = 0 + 0 = 0$, so $p+q \in \mathcal{U}$.

- $p \in \mathcal{U} \rightarrow cp \in \mathcal{U}$ for $c \in \mathbb{R}$. In the same way as before, $(cp)(t) = c \cdot p(t)$, so $(cp)(z) = c \cdot p(z) = c \cdot 0 = 0$, so $cp \in \mathcal{U}$.

b) Let $V = \{p \in \mathbb{P}_n : p(t) = a + t^2, a \in \mathbb{R}\} \subset \mathbb{P}_n$. Is V a subspace of $\mathbb{P}_n$? Why?

It's not because it doesn't satisfy the three previous conditions. For starters $0 \notin V$ since $p(t) = 0$ does not have the shape $p(t) = a + t^2$ even if $a = 0$.

④ $A = \begin{bmatrix} 2 & 2 & 1 \\ -7 & 4 & 2 \\ 5 & 0 & 0 \end{bmatrix}$

a) Find all the eigenvalues of A . Can we know from this if A is diagonalizable?

For that we solve $\det(A - \lambda Id) = 0$ for λ . So, $\det(A - \lambda Id) = 5 \begin{vmatrix} 2 & 1 \\ 4 - \lambda & 2 \end{vmatrix} - \lambda \begin{vmatrix} 2 & 2 \\ -7 & 4 - \lambda \end{vmatrix} =$

$$= 5(4 - 4 + \lambda) - \lambda((-2 - \lambda)(4 - \lambda) + 14) = 5\lambda - \lambda(-8 + 2\lambda - 4\lambda + \lambda^2 + 14) = 5\lambda - \lambda(\lambda^2 - 2\lambda + 6) =$$

$$= 5\lambda - \lambda^3 + 2\lambda^2 - 6\lambda = -\lambda^3 + 2\lambda^2 - \lambda = 0 \rightarrow \lambda(-\lambda^2 + 2\lambda - 1) = 0$$

$$\downarrow \quad \hookrightarrow \lambda = \frac{-2 \pm \sqrt{4 - 4}}{-2} = 1 \text{ (mult. 2)}$$

$$\lambda = 0$$

With this information alone we can't tell whether A is diagonalizable, we need to find the eigenspaces.

b) Give a basis for every eigenspace.

For that we need to find solutions of $(A - \lambda Id)x = 0$ for each eigenvalue λ .

$$\lambda = 0$$

$$\begin{bmatrix} -2 & 2 & 1 & 0 \\ -7 & 4 & 2 & 0 \\ 5 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} -2 & 2 & 1 & 0 \\ 0 & -6 & -3 & 0 \\ 0 & 10 & 5 & 0 \end{bmatrix} \sim \begin{bmatrix} -2 & 2 & 1 & 0 \\ 0 & 1 & 1/2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & -1/2 & 0 \\ 0 & 1 & 1/2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1/2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{cases} x_1 = 0 \\ x_2 = -\frac{x_3}{2} \end{cases}$$

So a basis is $\left\{ \begin{bmatrix} 0 \\ -1/2 \\ 1 \end{bmatrix} \right\}$.

Now $\lambda = 1$.

$$A - Id = \begin{bmatrix} -3 & 2 & 1 \\ -7 & 3 & 2 \\ 5 & 0 & -1 \end{bmatrix} \sim \begin{bmatrix} -3 & 2 & 1 \\ 0 & -5 & -1 \\ 0 & 10 & 2 \end{bmatrix} \sim \begin{bmatrix} -3 & 2 & 1 \\ 0 & 1 & 1/5 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1/5 \\ 0 & 1 & 1/5 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{cases} x_1 = \frac{1}{5}x_3 \\ x_2 = -\frac{1}{5}x_3 \end{cases}$$

So a basis is $\left\{ \begin{bmatrix} 1/5 \\ -1/5 \\ 1 \end{bmatrix} \right\}$.

c) Is A diagonalizable?

No, since the multiplicity of the eigenvalue 1 doesn't match the dimension of its eigenspace.

$$\textcircled{5} \quad v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

a) Use the basis to find an orthogonal basis $\{u_1, u_2, u_3\}$.

Since $\{v_1, v_2, v_3\}$ is not an orthogonal basis we can use Gram-Schmidt.

$$u_1 = v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$u_2 = v_2 - \frac{v_2 \cdot u_1}{u_1 \cdot u_1} u_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} - \frac{2}{3} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/3 \\ 1/3 \\ -2/3 \end{bmatrix}$$

$$u_3 = v_3 - \frac{v_3 \cdot u_1}{u_1 \cdot u_1} u_1 - \frac{v_3 \cdot u_2}{u_2 \cdot u_2} u_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} - \frac{1}{3} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \frac{1/3}{2/3} \begin{bmatrix} 1/3 \\ 1/3 \\ -2/3 \end{bmatrix} = \begin{bmatrix} 1/2 \\ -1/2 \\ 0 \end{bmatrix}$$

b) Let $y = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. Find the best approx. of y by vectors of $\text{Span}\{u_1, u_3\}$.

By the best approximation theorem, that is given by the orthogonal projection of y onto $\text{Span}\{u_1, u_3\}$, so

$$\hat{y} = \frac{y \cdot u_1}{u_1 \cdot u_1} u_1 + \frac{y \cdot u_3}{u_3 \cdot u_3} u_3 = \frac{2}{3} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \frac{1/2}{1/2} \begin{bmatrix} 1/2 \\ -1/2 \\ 0 \end{bmatrix} = \begin{bmatrix} 7/6 \\ 1/6 \\ 2/3 \end{bmatrix}$$

$\textcircled{6}$ True or false?

a) Diagonalizable $\rightarrow$ Different eigenvalues.

False, it's the other way around. Take for instance $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

b) $W \subset V$. If $u \neq 0$, $u \in W^\perp$, then $\|u\| = 0$.

False. By definition of norm $\|u\| = 0 \iff u = 0$.

c) A, B similar $\rightarrow \det A = \det B$.

True. A similar to B means $\exists P$ invertible s.t. $A = PBP^{-1}$, so $\det A = \det(PBP^{-1}) = \det P \cdot \det B \cdot \det P^{-1} = \det P \cdot \det P^{-1} \cdot \det B = \det B$ (because $\det P \cdot \det P^{-1} = \det(PP^{-1}) = \det Id = 1$, or $\det P^{-1} = \frac{1}{\det P}$).

⑦ What is an eigenvector of a sudoku? And its eigenvalue?

The eigenvector is $(1, 1, 1, 1, 1, 1, 1, 1, 1)$ with eigenvalue $1+2+3+4+\dots+9 = 45$, since it's the sum of all the entries in any row of a sudoku.